Ch 6.3: Dimension Reduction

Lecture 16 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Mon, Mar 14, 2022

Announcements

Last time:

- Midterm
- Grades posted by Wednesday

This lecture:

- PCA / PCR
- PLS

Announcements:

- Homework due Friday
  - Monday 3/13: 6.3 Dimension Reduction
  - Wednesday 3/15: ?? 6.4
- Friday lab 6.5.2-3

Section 1

Last time

Shrinkage

Find β to minimize

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

subject to:

Least Squares:

No constraints

Ridge:

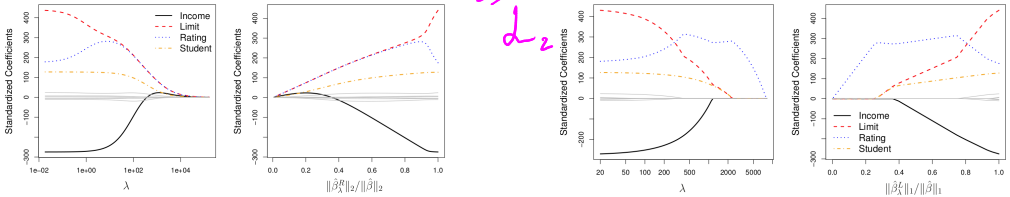
$$\sum_{j=1}^{p}\beta_j^2 \leq s$$

ℓ_2

The Lasso:

$$\sum_{j=1}^{p}|\beta_j| \leq s$$

ℓ_1



Section 2

Dimension Reduction

Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, \cancel{X}_M \; Z$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

m

$p \gg M$

Coeff

Find "good" ones

An example or two

$p = 3$ $\quad X_1, X_2, X_3$ $\quad Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$

$\varphi_{1,1} = 1$

$Z_1 = X_1 + 0X_2 + 0X_3$

$\varphi_{2,2} = 1/2$ $\quad \varphi_{3,2} = \frac{1}{2}$

$Z_2 = 0 \cdot X_1 + \frac{1}{2} X_2 + \frac{1}{2} X_3$

$\varphi_{1,2} = 0$

The goal

$(z_{i1}, \ldots, z_{iM})$

$\downarrow$

(z_i, y_i) transformed data

- Find good φ's
- Fit regression model on Z_i's using least squares

learn these coeff (like $\hat{\beta}_i$)

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

z_i

PCA - finding φ

$\hookrightarrow$ PCR = PCA + Regression

Dimension Reduction:

p = big $\rightsquigarrow$ M smaller

Section 3

PCA

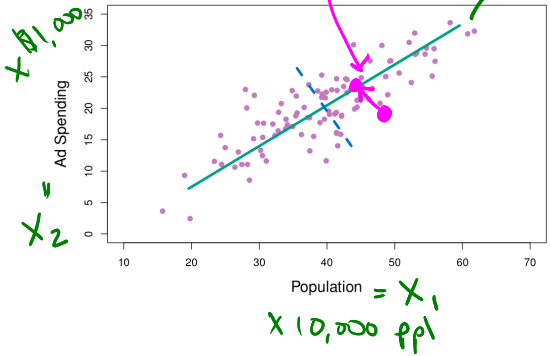
Principal Component Analysis

An example dataset

projection onto the line

1st Principal Component

Direction of most variability in the data



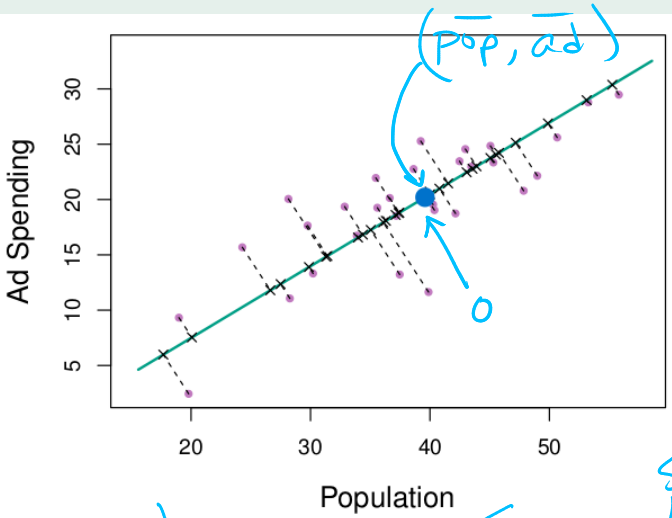
If project data onto the line and take variance this line has the higest variance.

Y = sales

Projection onto a line

https://www.desmos.com/calculator/cih7wy8oyg

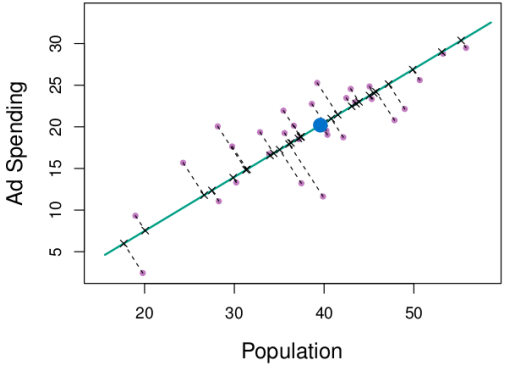
Projection onto first PC



score

$$Z_1 = 0.839 \cdot (\text{pop} - \overline{\text{pop}}) + 0.544 \cdot (\text{ad} - \overline{\text{ad}})$$

What does it mean to have the highest variance



of every linear combo of pop, ad

this line has the highest variance of

$$Var(\varphi_{1,1}(pop - \bar{pop}) + \varphi_{2,1}(ad - \bar{ad}))$$

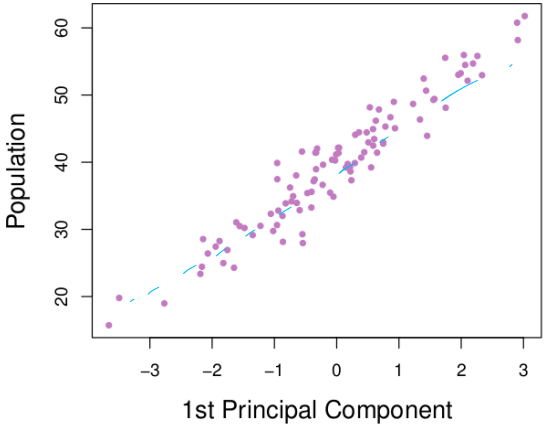
force $\varphi_{11}^2 + \varphi_{21}^2 = 1$

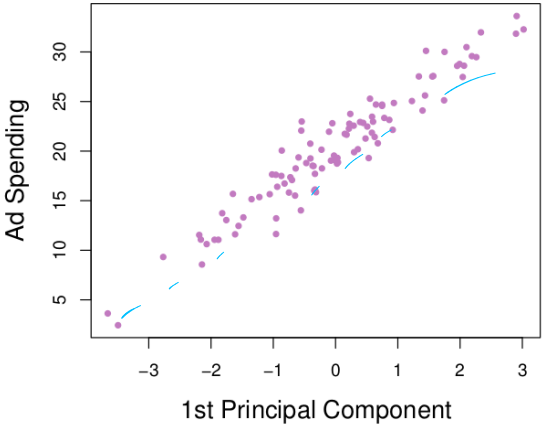
Prevents blow up of variance

Toy for learning PCA

https://www.desmos.com/calculator/qq14tyjz0z

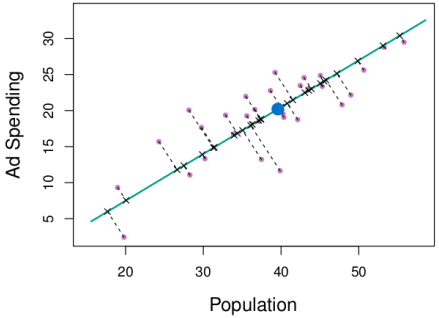
Principal component scores

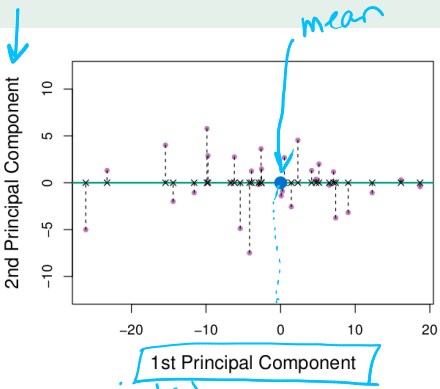




$$z_{i1} = 0.839 \cdot (\mathrm{pop}_i - \overline{\mathrm{pop}}) + 0.544 \cdot (\mathrm{ad}_i - \overline{\mathrm{ad}})$$

Another view





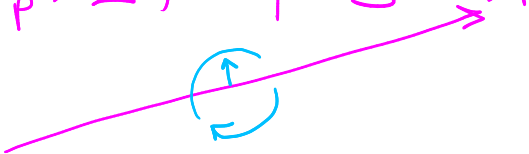
The score can be viewed as projected relation to the mean

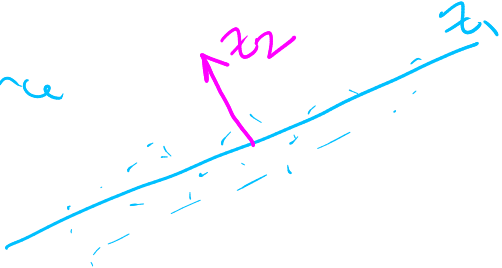
$z_{i1} < 0 \Rightarrow$ below avg pop size and below avg spending

The other principal components

z_1
z_2
$\vdots$
z_M

- You already have $z_1 = \sum \varphi_{i1} x_i$
- z_2 is linear combination of variables (x_i) uncorrelated to z_1
  - ↳ Find one w/ most variance
- If $p > 2$, $p = 3$





Section 4

Principal Components Regression

So you've found your PCA coefficients

Now what?

We want to do regression

Do on $Z_1, \ldots, Z_M$

Book calls

"Principal component regression"

What are we assuming?

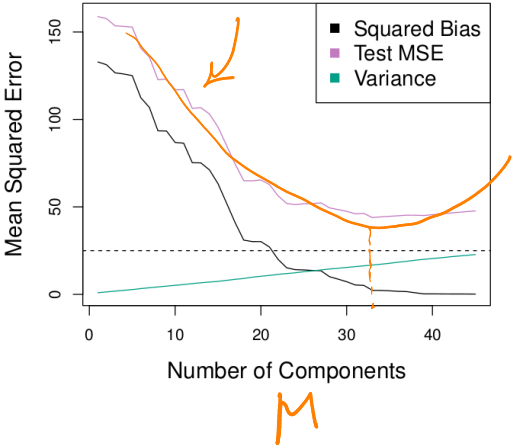
The directions in which X_i's have the most variation are also the ones that are associated to Y

If the assumption holds fit on Z_i better than X_i since fewer $\Rightarrow$ less overfitting

Doing better

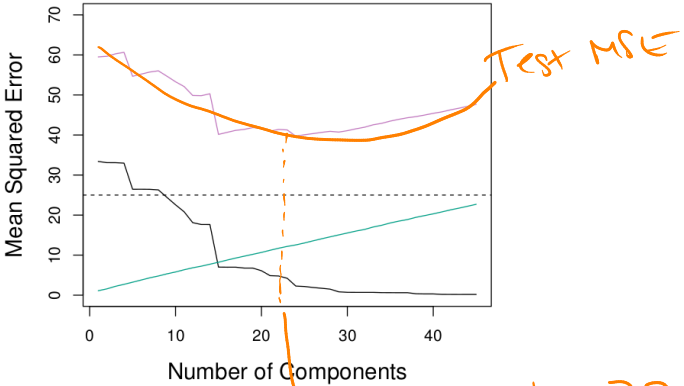
Example with simulated data: $n = 50$ observations of $p = 45$ predictors

Y is a function of 2 predictors



Doing better

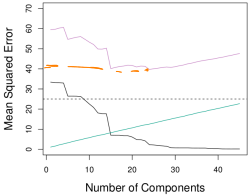
Example with simulated data: $n = 50$ observations of $p = 45$ predictors
Y is a function of all predictors



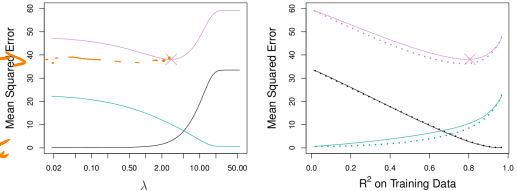
Comparison to results on shrinkage

Y is a function of all predictors

PCA



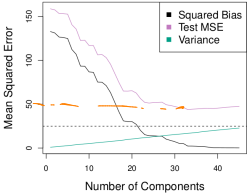
best test MSE



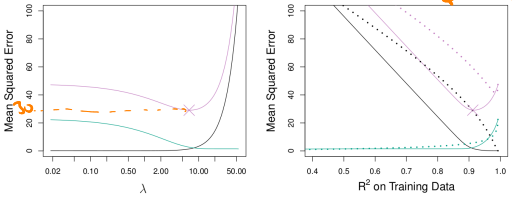
Ridge

Y is a function of 2 predictors

PCA

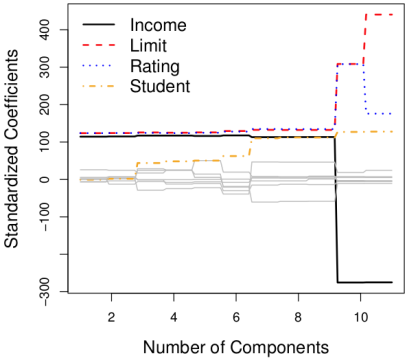


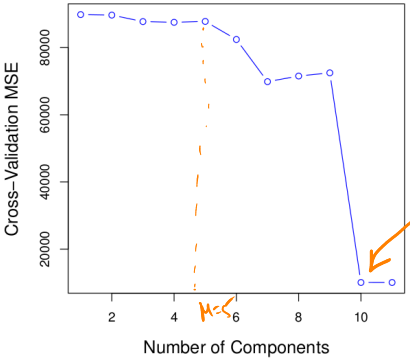
Ridge better



Ridge

Picking M





Best
M=10

Not a huge gain
p=11

Properties of PCR

- PCR does better when info is contained in the first few PC's
- Big warning: This is not variable selection!
  - ↳ PC's are linear combos of input data
    Not original variables themselves
- # of PC's can be chosen w/ CV method
- PCA is many other applications

Section 5

Partial Least Squares

Supervised alternative

PCR: Non-supervised

- Did not use Y before learning the coeff for Z_i's
- No guarantee that the directions Z_i explain Y

Partial Least Squares (PLS):

- Identify new features $Z_1, \cdots, Z_M$ linear combos of original where quality measure involves Y
- Fit linear model using least squares on these M features

First direction Z_1 for Partial Least Squares (PLS)

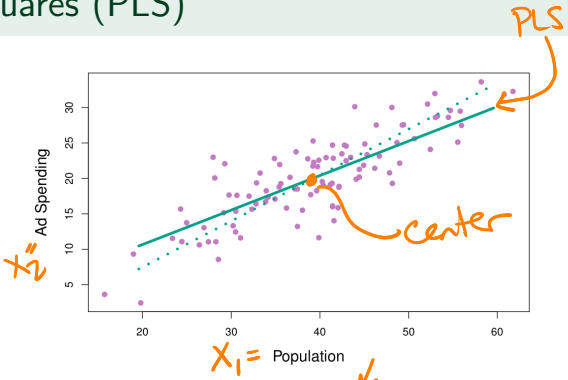
Lin Reg $(X_1, Y) \rightsquigarrow \hat{\beta}_1$

— " — $(X_2, Y) \rightsquigarrow \hat{\beta}_2$

$(X_3, Y \rightsquigarrow \hat{\beta}_3$

- Set φ_{j1} equal to the coefficient from simple linear regression of Y onto X_j
- The first direction is

$$Z_1 = \sum_{j=1}^{p} \varphi_{j1} X_j$$



Ex. Prediction of Y =Sales (not shown) on X_1 =Population and X_2 =Ad Spending

- Solid green: First PLS direction
- Dashed: First PC direction

Ex) PLS chose direction less change in ad spending than PC, less involved w/ predicting sales

Second (and more) PLS directions

- Regress each variable on Z_1 and take residuals
- Compute Z_2 using *orthogonalized data* same as for Z_1

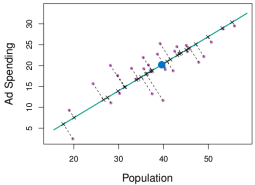
Rinse + repeat

- # of predictors
  Can be chosen w/ CV
- Not commonly used
  Mixed bag in results vs PCR

TL;DR

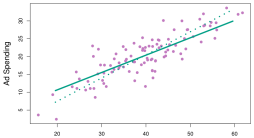
PCA

- Unsupervised dimensionality reduction
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



PLS

- Supervised dimensionality reduction
- Choose component Z_1 by using simple regression coefficients of each X_i onto Y
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained
- Not a particular benefit, so usually default to PCA unless you have a good reason for this



Next time

Next time:

- Ch 6.4 Issues in High Dimensions

This week:

- Mon 3/14: 6.3 PCA and regression
- Weds 3/16: 6.4 High Dimensions
- Fri 3/18: 6.5 Lab
  - Homework #7 Due
  - Covers Monday and Wednesday only

A Bit short, likely could add more content/examples. Maybe use PCA for clustering too? More real data!